The socle case with $L \subsetneq \mathrm{Aut}(D)$:
a tabulation of the $|G| = 16$ three-way lock and the correction constant $c_{[\nu],\sigma}$ for $H = \mathbb{Z}/2$

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Provenance

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Recipient Rick (co-developer of the "$\operatorname{im}\varphi^+ = \ker o_\nu$" reformulation; peer review). CC internal: Neil Ghani / Robin Langer.

Purpose Per Rick's request (email 2026-10-01): the socle case with the $L \subsetneq \mathrm{Aut}(D)$ tabulation — the $|G| = 16$ witness showing the three-way lock and the correction constant $c_{[\nu],\sigma}$ for $H = \mathbb{Z}/2$.

Corresponding WIP commit `648fa09125f07bad7ea923655c6ffbf63bb305e2` (repo `github.com/scotmacbeth/work-in-progress`, `origin/main`; log: "*o_ν non-trivial-kernel: affine over $(D,\circ)$, torsor needs $L \subsetneq \mathrm{Aut}(D)$ — corrected* `o-nu-affine-torsor` *claim (computed, 16/16); trustcheck GREEN*"). *Note:* this commit records the correction; the two full proof `.tex` sources transcribed below (`2026-09-30-general-nontrivial-onu.tex`, `2026-09-30-general-H-onu-offset.tex`) are later local additions not yet in that push.

What this is. A faithful tabulation assembled from three existing proof files. It introduces *no* new mathematics; every line carries its honest trust grade (`proved` / `computed`) exactly as recorded in the sources:

[S1] `2026-09-29-solvability-obstruction-o-nu.tex` — the obstruction $o_\nu : H^2_+ \to C^3/A(Z^2_\circ)$, $\operatorname{im}\varphi^+ = \ker o_\nu$ (`proved`, trivial kernel); the $|G| = 16$ relation and torsor (there `computed`).

[S2] `2026-09-30-general-nontrivial-onu.tex` — the non-trivial-kernel home $(D,\circ)$/twist $\widehat{\sigma}$ correction and the by-hand three-way lock (`proved`, $H = \mathbb{Z}/2$, 112/112).

[S3] `2026-09-30-general-H-onu-offset.tex` — the general-H offset $c^d = d_+[\delta_\bullet(d)]$ and the torsor criterion (`proved` general H for the offset/criterion; main biconditional `computed` 34/34).

1 Setup: the extension and the holomorph witness

Fix the skew-brace extension

$$0 \longrightarrow D \longrightarrow G \longrightarrow H \longrightarrow 0, \qquad H = \mathbb{Z}/2, \quad D = \mathbb{Z}/8,$$

with D a *non-trivial* brace ideal: $(D,+)=\mathbb{Z}/8$ carries the brace action $\lambda^D_x=5^x$, so $d\circ e=d+\lambda^D_d(e)$ and $(D,\circ)$ is abelian. A normalised set-section $s:H\to G$, $s(0)=0$, gives the good triplet [S2, §1]

$$\nu_h=\lambda_{s(h)}|_D,\qquad \mu_h(x)=-s(h)+x+s(h),\qquad \widehat{\sigma}_h(x)=s(h)^{-1}\circ x\circ s(h).$$

In all witnesses $(G,+)$ is abelian, so $\mu=\mathrm{id}$. Here $\widehat{\sigma}$ is the **$\circ$-conjugation**, the correct twist (§4); it replaces the additive σ_+ used in [S1].

The inner group L and the holomorph. Write $L:=\mathrm{im}(\lambda^D|_D)=\{\lambda^D_d : d\in D\}\le\mathrm{Aut}(D,+)$. For $D=\mathbb{Z}/8$, $\mathrm{Aut}(\mathbb{Z}/8)=\{1,3,5,7\}\cong(\mathbb{Z}/8)^\times$, and since $\lambda^D_1=5$ generates $\{1,5\}$,

$$\boxed{L=\{1,5\}\subsetneq\mathrm{Aut}(\mathbb{Z}/8)\,.}$$

Realising G as a regular subgroup of the holomorph $\mathrm{Hol}(D)=D\rtimes\mathrm{Aut}(D)$ with $|G|=|D|\cdot|H|=16$ yields the family tabulated below; the enumeration source is the independent regular-subgroup-of-holomorph computation [S1, Prop. 5; `2026-09-29-o-nu-kernel-vs-image.py`]. The subgroup $\{1,5\}$ is the *socle coset* in $\mathrm{Aut}(\mathbb{Z}/8)$ (the squares); it is the home of $\nu\widehat{\sigma}$ when no torsor arises (§6).

2 The socle statement, transcribed honestly

Rick asked specifically about the socle. The record must be stated carefully because *two distinct* "socle" notions appear, and one earlier claim was **refuted**.

Correction (honest): socle is NOT the discriminator

An earlier gloss asserted "*separation requires $D\not\subseteq\mathrm{Soc}(G)$*" (the old `rem:socle` / summary T3). **This is FALSE.** Rick's referee point, confirmed in MacBeth's write ledger [`scratch/write-2026-09-26.md`, item 1], is that the witness brace **sb#1 has $D=\mathrm{Soc}(G)$ exactly, yet still separates** (Φ non-injective on W_4, $(G,+)=\mathbb{Z}/2\times\mathbb{Z}/4$ [`2026-09-25-soc-w4-check.py`, `2026-09-25-lean-w4-noninjective.md`]). Hence socle-containment is **not** a necessary condition for separation and must not be stated as a theorem.

The "socle" that *does* control the torsor dichotomy below is a different object: it is membership of $\nu\widehat{\sigma}$ in the socle coset $L=\{1,5\}$ of $\mathrm{Aut}(D)$, i.e. whether $\nu\widehat{\sigma}\equiv 1 \pmod 4$. Calling $\nu\widehat{\sigma}$ "socle" / "non-socle" in §5–§6 refers to this coset, *not* to $D\subseteq\mathrm{Soc}(G)$.

3 The three-way lock

Theorem 1 (Three-way lock; [S2, Thm 3.8], `proved` ($H=\mathbb{Z}/2$))**.** *For every $x\in D$,*

$$(\lambda^D_x-1)\,b\;=\;2\,\delta_1(x)\;=\;2\big(\nu_1\widehat{\sigma}_1(x)-x\big)\quad \textit{in } (D,+),\qquad b:=\beta(1,1),\quad u:=s(1),\quad b=2u,$$

where $\delta_h:=\nu_h\widehat{\sigma}_h-\mathrm{id}$ is the intrinsic defect [S2, Lem. 3.1; $\delta_h(x)=\lambda_x(s(h))-s(h)$, = RY (3.1), general, no triviality]. At x a generator, $(\lambda^D_1-1)b\equiv 2(\nu\widehat{\sigma}-1) \pmod{|D|}$; for $D=\mathbb{Z}/8$ ($\lambda^D_1=5$),

$$\boxed{4b\;\equiv\;2(\nu\widehat{\sigma}-1)\pmod 8\,,}\qquad [\beta]\neq 0\ (b\ \textit{odd})\iff \nu\widehat{\sigma}\equiv 3 \pmod 4.$$

The derivation is one skew-brace-axiom instance $a\circ(b'+c')=(a\circ b')-a+(a\circ c')$ at $a=x\in D$, $b'=c'=u$ [S2, proof of Thm 3.8]. The general-H parent is the kernel coupling ($\star$) [S3, Lem. 2.1]

$$(\lambda^D_d-\mathrm{id})\,\beta(k,k')=\delta_k(d)+\delta_{k'}(d)-\delta_{k+_Hk'}(d),$$

of which the lock is the $H=\mathbb{Z}/2$, $k=k'=1$ slice ($\delta_0=0$). ($\star$) is `proved` for general H.

Verified cases, with grades. Exactly as recorded in the sources:

Statement	Range	Count	Grade
Lock $(\lambda_x^D - 1)b = 2\delta_1(x)$ by hand	$H = \mathbb{Z}/2$, all nontriv. kernels	112/112	`proved` [S2]
— 8 on $\mathbb{Z}/16$ ($[\beta] \neq 0$)	$D = \mathbb{Z}/8$		
— 96 on $\mathbb{Z}/2 \times \mathbb{Z}/8$ ($[\beta] = 0$)	$D = \mathbb{Z}/8$		
— 8 on $\mathbb{Z}/4 \times \mathbb{Z}/2$	$D = \mathbb{Z}/4$		
$\lvert G\rvert = 16$ relation $4b \equiv 2(\nu\sigma - 1)$	$D = \mathbb{Z}/8$ family	24/24	`computed` [S1]
— 8 on $\mathbb{Z}/16$; 16 on $\mathbb{Z}/2 \times \mathbb{Z}/8$			
Twist $\widehat{\sigma}$ beats additive σ_+	coupling constant	104/104 vs 80/104	`proved` [S2]
$(D, +)$ mult. differential fails $d^2 = 0$	good triplets, $D = \mathbb{Z}/8$	2/16	`computed` [S2]
Operator-level assembly exact	genuine nontriv. braces	33/33	`computed` [S2]
Abstract = genuine $\operatorname{im}\varphi^+$	all triplets $H = \mathbb{Z}/2$	16/16	`computed` [S2]

The 104/104 vs 80/104 split isolates the twist: the 24 failures of σ_+ are the $D = \mathbb{Z}/2 \times \mathbb{Z}/8$ braces with $\nu \not\equiv 1 \pmod 4$ [S2, Thm 3.5].

4 The correction constant $c_{[\nu],\sigma}$

Definition 2 (Correction constant / affine offset)**.** For $H = \mathbb{Z}/2$ the obstruction is the affine map $o_\nu([\beta]) = (\lambda_1^D - 1)b - 2(\nu\widehat{\sigma} - 1)$ on $H_+^2 = D/2D$, with constant offset

$$\boxed{c_{[\nu],\sigma} \;=\; 2(\nu\widehat{\sigma} - 1)} \qquad (\text{for } D = \mathbb{Z}/8 :\; c \in \{0, 4\}).$$

Provenance of the constant, honestly.

- In [S1, §§6–7] this constant is written $c = 2(\nu\sigma - 1)$ with the *additive* σ, and both the $\lvert G\rvert = 16$ relation and the torsor statement are graded `computed` / **open** for a general proof.

S2, Thm 3.5 **corrects** $\sigma \mapsto \widehat{\sigma}$ (the $\circ$-conjugation): the constant matching genuine braces is $2(\nu\widehat{\sigma} - 1)$, 104/104 vs 80/104 for σ_+. With the corrected twist the lock and torsor are upgraded to `proved` at $H = \mathbb{Z}/2$.

S3, Cor. 2.3 gives the general-H closed form: the offset is the additive coboundary of the defect section,

$$c^d \;=\; d_+[\,\delta_\bullet(d)\,] \;=\; d_+[\,k \mapsto \delta_k(d)\,] \qquad (\forall d \in D),$$

`proved` for general H. At $H = \mathbb{Z}/2$ this is $c^d = 2\delta_1(d) = 2(\nu\widehat{\sigma} - 1)d$, recovering $c_{[\nu],\sigma}$.

Tabulation of $c_{[\nu],\sigma}$ for the $\lvert G\rvert = 16$ witness ($D = \mathbb{Z}/8$, $L = \{1, 5\}$).

$\nu\widehat{\sigma} \bmod 4$	socle coset?	$c = 2(\nu\widehat{\sigma} - 1)$	solvable b ($4b \equiv c$)	$[\beta]$	$(G, +)$
$\equiv 1$	yes ($\nu\widehat{\sigma} \in \{1, 5\}$)	0	$\{0, 2, 4, 6\} \ni 0$	$[\beta] = 0$ forced	$\mathbb{Z}/2 \times \mathbb{Z}/8$
$\equiv 3$	no ($\nu\widehat{\sigma} \in \{3, 7\}$)	4	$\{1, 3, 5, 7\} \not\ni 0$	$[\beta] \neq 0$ forced	$\mathbb{Z}/16$

(Grade: the relation $4b \equiv c$ is `proved` at $H = \mathbb{Z}/2$ via the lock [S2]; the enumeration match is `computed` 24/24 [S1].) The realisable set of b is the coset $\{b : 4b \equiv c\}$ of $\ker(\times 4) = \{0, 2, 4, 6\}$.

5 The torsor dichotomy

Proposition 3 (Torsor dichotomy; [S2, Prop. 5.2], [S3, Thm 4.1])**.** $\operatorname{im}\varphi^+ = o_\nu^{-1}(0)$ *is an affine coset of* $\ker(\lambda_1^D - 1)$*. It is a* torsor without basepoint *(*$0 \notin \operatorname{im}\varphi^+$*) iff the offset* $c_{[\nu],\sigma} = 2(\nu\widehat{\sigma} - 1)$ *is nonzero in the target, equivalently iff* $L = \operatorname{im}(\lambda^D) \subsetneq \operatorname{Aut}(D)$ *and* $\nu\widehat{\sigma}$ *is* non-socle*. General-H criterion [S3]:* $0 \in \operatorname{im}\varphi^+$ *iff, for every* $d \in D$*, the defect section* $\delta_\bullet(d) : (H, +) \to (D, +)$ *is a group homomorphism (i.e.* $c^d = 0$*).* Grade: `proved` *for general H (offset/criterion);* `proved` *(*$H = \mathbb{Z}/2$*) for the dichotomy; main biconditional* `computed` 34/34 *triplets (*$H = \mathbb{Z}/2, \mathbb{Z}/3$*;* 86 *genuine braces,* $(\star)$ *holding* 86/86*).*

Kernel D	$L = \operatorname{im}(\lambda^D)$	$L \subsetneq \operatorname{Aut}(D)$?	offset c	$\operatorname{im}\varphi^+$	verdict
$\mathbb{Z}/8$ (non-triv. $\lambda_1^D = 5$)	$\{1, 5\}$	**yes**	0 or 4	coset of $\{0, 2, 4, 6\}$	**torsor** at $\nu\widehat{\sigma} \equiv 3$ (no basep
$\mathbb{Z}/4$ (non-triv.)	$\operatorname{Aut}(\mathbb{Z}/4)$	no ($L = \operatorname{Aut}$)	$\equiv 0$	$\{0\}$	based; $b \equiv 0$ always; **no to**

The contrast with $\mathbb{Z}/4$**.** The non-trivial brace on $D = \mathbb{Z}/4$ has $L = \operatorname{Aut}(\mathbb{Z}/4)$ (the full automorphism group). Then $(\lambda_1^D - 1)$ is surjective enough that $2(\nu\widehat{\sigma} - 1) \equiv 0$ is forced, so $c \equiv 0$, $b \equiv 0$ always, and $\operatorname{im}\varphi^+ = \{0\}$: a *based* subgroup, no torsor [S2, Prop. 5.2]. This is the essential point for Rick: *the torsor-without-basepoint phenomenon requires* $L \subsetneq \operatorname{Aut}(D)$; when $L = \operatorname{Aut}(D)$ the obstruction is linear and 0 is always realisable. In the trivial-kernel (RY) regime o_ν is likewise linear [S1, §7], so the torsor is a strictly non-trivial-kernel effect.

Remark 4 (Cohomological necessity is strictly weaker)**.** $(\lambda_d^D - \mathrm{id})_*[\beta] = 0$ in H^2_+ is necessary for realisability but strictly weaker than $(\star)$: the offset c^d must be matched at the *cochain* level by a single representative. At $D = \mathbb{Z}/8$, $H = \mathbb{Z}/2$ the cohomology condition admits both classes while $\operatorname{im}\varphi^+$ is a single class — exactly the torsor phenomenon [S3, Cor. 2.4, `2026-09-30-coho-star.py`, `computed`].

Summary for Rick

- **Witness.** $0 \to \mathbb{Z}/8 \to G \to \mathbb{Z}/2 \to 0$, $\lambda_x^D = 5^x$, $|G| = 16$, $L = \{1, 5\} \subsetneq \operatorname{Aut}(\mathbb{Z}/8)$ (holomorph construction). `computed` [S1].
- **Socle, honest.** "Separation requires $D \not\subseteq \operatorname{Soc}(G)$" is **refuted** (sb#1 has $D = \operatorname{Soc}(G)$ yet separates); socle-containment is **not** the discriminator. The "socle" that governs the torsor is the coset $L = \{1, 5\}$: $\nu\widehat{\sigma}$ socle $\Leftrightarrow \nu\widehat{\sigma} \equiv 1 \pmod 4$.
- **Three-way lock.** $4b \equiv 2(\nu\widehat{\sigma} - 1) \pmod 8$; `proved` by hand $H = \mathbb{Z}/2$, 112/112; $|G| = 16$ relation `computed` 24/24. Twist must be $\widehat{\sigma}$ (104/104 vs 80/104).
- **Correction constant.** $c_{[\nu],\sigma} = 2(\nu\widehat{\sigma} - 1)$; general-$H$ closed form $c^d = d_+[\delta_\bullet(d)]$ `proved`.
- **Torsor dichotomy.** Torsor without basepoint $\Leftrightarrow L \subsetneq \operatorname{Aut}(D)$ and $\nu\widehat{\sigma}$ non-socle; $\mathbb{Z}/4$ ($L = \operatorname{Aut}$) is based, no torsor. `proved` ($H = \mathbb{Z}/2$); general-H criterion `proved`; main biconditional `computed` 34/34.

Credit. The reformulation "$\operatorname{im}\varphi^+ = \ker o_\nu$" is Rick's Remark 4 (referee report, 2026-09-26). This tabulation transcribes MacBeth's follow-up proofs [S1–S3] without alteration.